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ME 362
ME 362 VIBRATIONS I

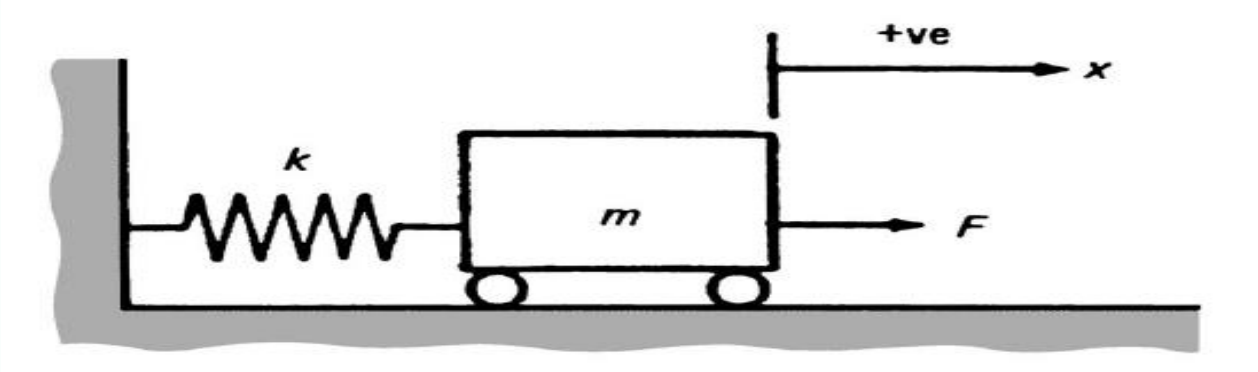
Lecture 2

RESPONSE TO HARMONIC EXCITATIONS

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FORCED UNDAMPED VIBRATION



$$m\ddot{x} + kx = F_o sin\omega t$$

$$\ddot{x} + \omega_n{}^2 x = (F_o/m) sin\omega t$$

$$x = x_h + x_p$$

$$x_h = A sin\omega_n t + B cos\omega_n t$$

$$x_p = X_o sin\omega t$$

$$\dot{x}_p = X_o \omega cos\omega t;\ \ddot{x}_p = -X_o \omega^2 sin\omega t$$

$$-X_o \omega^2 sin\omega t + X_o \omega_n{}^2 sin\omega t = (F_o/m) sin\omega t$$

$$X_o = \frac{(F_o/m)}{\omega_n{}^2 - \omega^2}$$

$$X_o = \frac{(F_o/k)}{(m/k)(\omega_n{}^2 - \omega^2)} = \frac{\delta}{1-\tau^2}$$

$$\frac{X_o}{\delta} = \frac{1}{1-\tau^2} = M$$

- The quantity X_o/δ represents the ratio of the dynamic to the static amplitude of motion and is called the ***magnification factor, amplification factor*, or *amplitude ratio***.
- The frequency ratio $\tau = \omega/\omega_n$.

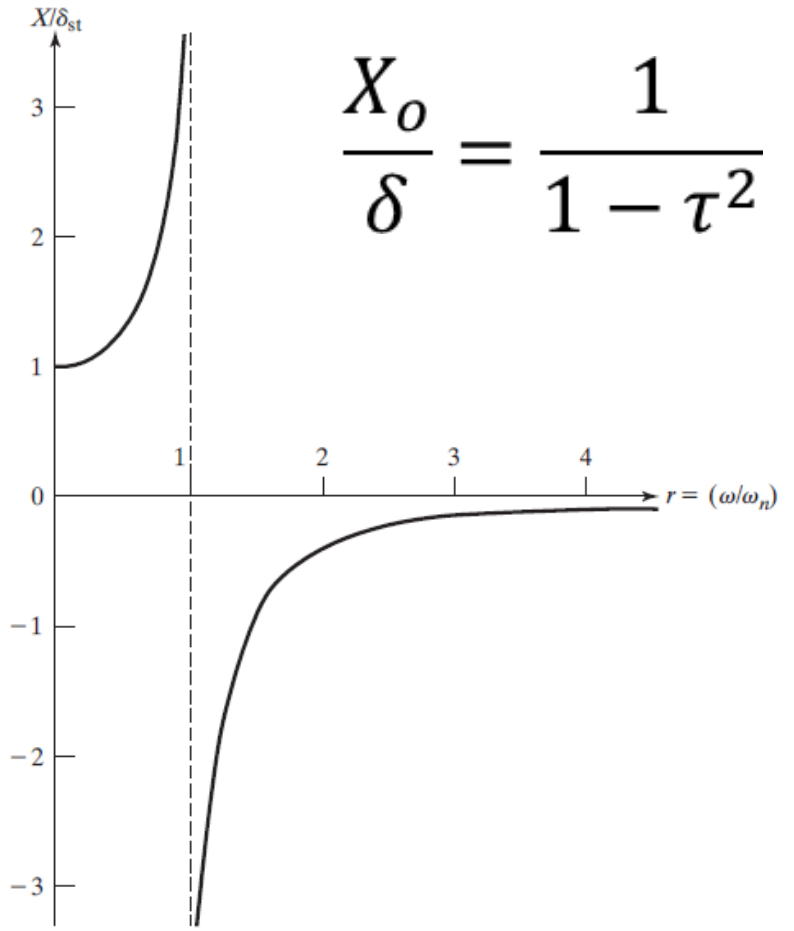
The total response then is,

$$x = A sin\omega_n t + B cos\omega_n t + \frac{\delta}{1-\tau^2} sin\omega t$$

FORCED UNDAMPED VIBRATION

- The response of the system can be identified to be of three types.
- When **0 ≤ τ < 1**; the denominator is positive and the response is Xsinωt. The harmonic response of the system is said to be in phase with the external force.
- When **τ > 1**; the denominator is negative, and the solution is –Xsinωt. The response is said to be 180° out of phase with the external force. Further, as $\tau \rightarrow \infty$, $X \rightarrow 0$ Thus the response of the system to a harmonic force of very high frequency is close to zero.
- When **τ =1**; the amplitude *X* becomes infinite. This condition, for which the forcing frequency is equal to the natural frequency of the system is called *resonance.*

$$\frac{X_o}{\delta} = \frac{1}{1-\tau^2}$$



Example 2-1

Compute and plot the response of a sound system subjected to a force of magnitude 23 N, driving frequency of twice the natural frequency and initial conditions given by x_0 = 0 m and v_0= 0.2 m/s. The mass of the system is 10 kg and the spring stiffness is 1000 N/m.

Solution 2–1

First compute the various coefficients for the response

$$\omega_n = \sqrt{\frac{1000\ \mathrm{N/m}}{10\ \mathrm{kg}}} = 10\,\frac{\mathrm{rad}}{\mathrm{s}}, \omega = 2\omega_n = 20\mathrm{rad/s},$$

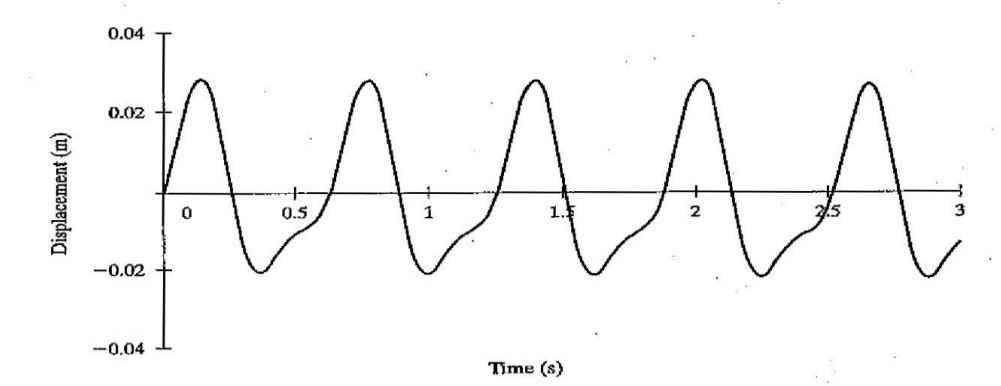
$$f_0 = \frac{23}{10} = 2.3\ \mathrm{N/kg}$$

$$\frac{v_0}{\omega_n} = \frac{0.2\ \mathrm{m/s}}{10\ \mathrm{rad/s}} = 20\ \mathrm{mm}$$

$$\frac{f_0}{\omega_n^2 - \omega^2} = \frac{2.3\ \mathrm{N/kg}}{(10^2 - 20^2)\mathrm{rad}^2/s^2} = -7.967\ mm$$

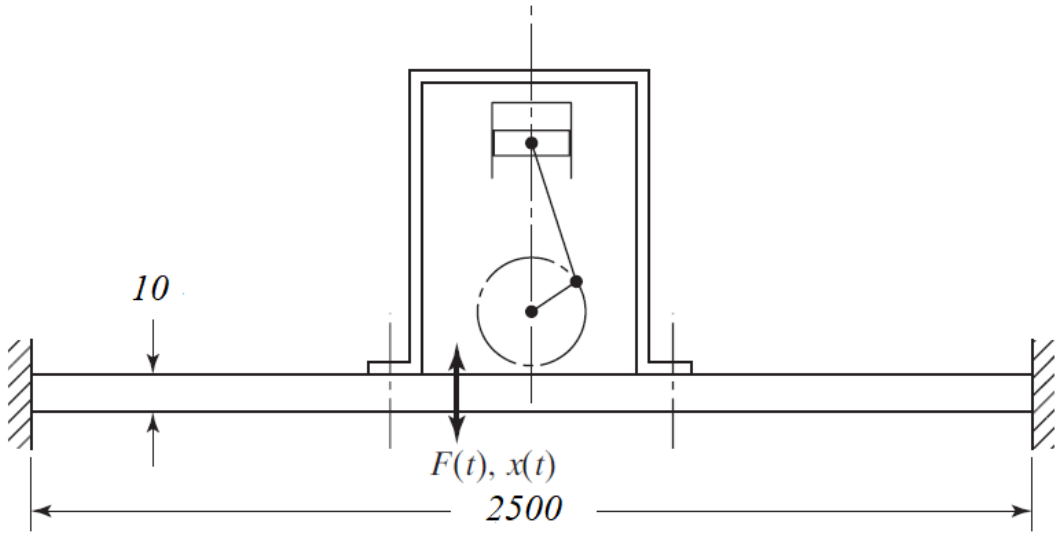
With these values, the response becomes

$$x(t) = 20\sin 10t + 7.967(\cos 10t - \cos 20t)\mathrm{mm}$$



Example 2-2

A reciprocating pump, weighing 50 kg, is mounted at the middle of a steel plate of thickness 10 mm, width 500 mm, and length 2500 mm, clamped along two edges as shown. During operation of the pump, the plate is subjected to a harmonic Primary force, F(t)=20 cos 6.28 *t* N. Find the amplitude of vibration of the plate. E=200 MPa



Solution: The plate can be modeled as a fixed-fixed beam.

The area moment of inertia

$$I = \frac{0.5 \times 0.01^3}{12} = 4.17 \times 10^{-8} m^4$$

And the bending stiffness of the beam is given by

$$k = \frac{192EI}{l^3} = \frac{192 \times 2 \times 10^8 \times 4.17 \times 10^{-8}}{2.5^3}$$

$$k = 102.48 \text{ N/m}$$

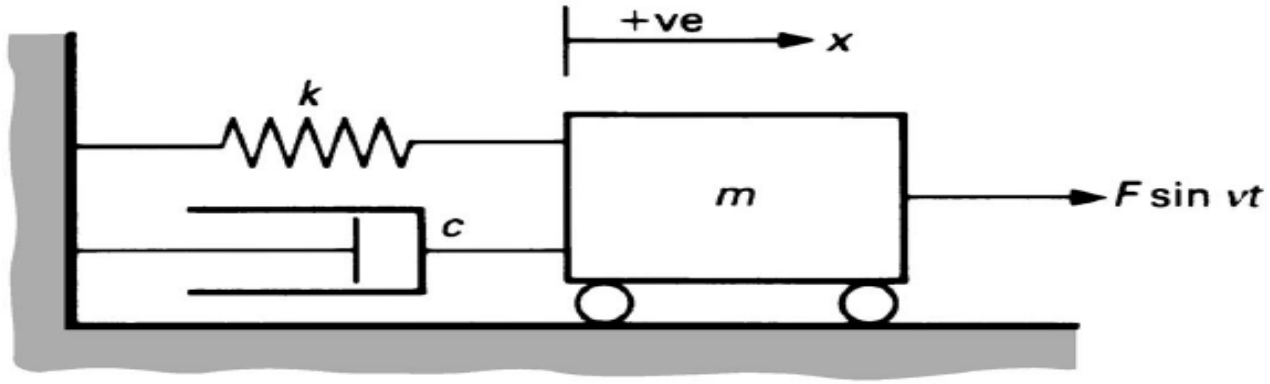
The amplitude is given by

$$X_o = \frac{(F_o/m)}{{\omega_n}^2 - \omega^2}$$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{20.496} = 4.53\, rad/s$$

$$X_o = \frac{(F_o/m)}{{\omega_n}^2 - \omega^2} = \frac{0.4}{18.92} = 21.14\, mm$$

FORCED DAMPED SYSTEMS



$$m\ddot{x} + c\dot{x} + kx = F_o sinvt$$

$$x = x_h + x_p$$

The homogenous solution, x_h depends on the value of ξ

The particular solution

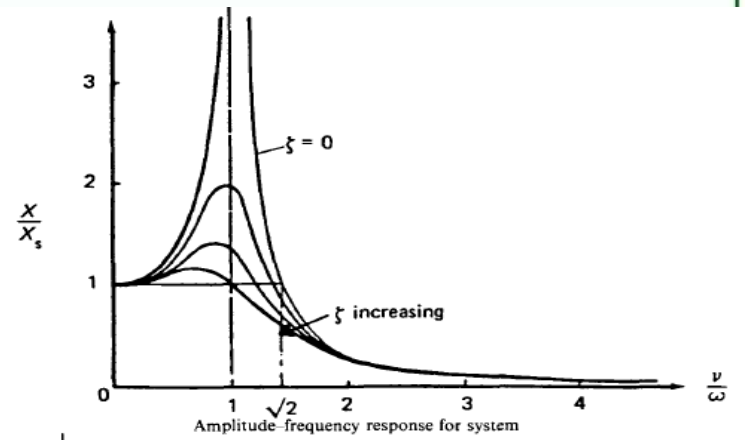
$$x_p = X_o \sin(\omega t - \emptyset)$$

$$\dot{x}_p = X_o \omega cos(\omega t - \emptyset); \quad \ddot{x}_p = -X_o \omega^2 \sin(\omega t - \emptyset)$$

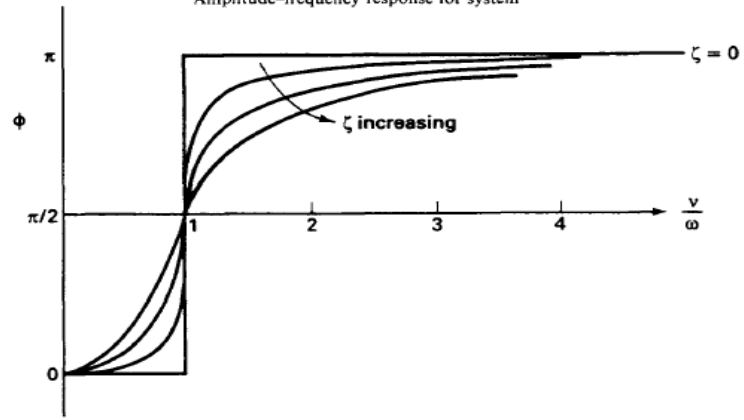
$$X_o = \frac{\mathrm{F_o}/m}{\sqrt{(\omega_\mathrm{n}^2 - \omega^2)^2 + (2\xi\omega_\mathrm{n}\omega)^2}} = \frac{\delta}{\sqrt{(1-\tau^2)^2 + (2\xi\tau)^2}};$$

$$\tau = \omega/\omega_n$$

$$\emptyset = \tan^{-1}\left(\frac{2\xi\omega_\mathrm{n}\omega}{\omega_\mathrm{n}^2 - \omega^2}\right) = \tan^{-1}\frac{(2\xi\tau)}{(1-\tau^2)}$$



Amplitude–frequency response for system



Phase–frequency response for system

Amplitude and Phase response

$$\frac{X_o}{\delta} = M = \frac{1}{\sqrt{(1-\tau^2)^2 + (2\xi\tau)^2}}$$

$$@\ \tau = 0, M = 1;\ \text{as}\ \tau \to 1\ \&\ \xi = 0; M \to \infty$$

$$\text{as}\ \tau \to \infty, M \to 0; \text{as}\ \tau \to 1\ \&\ \xi = 0; M \to \infty$$

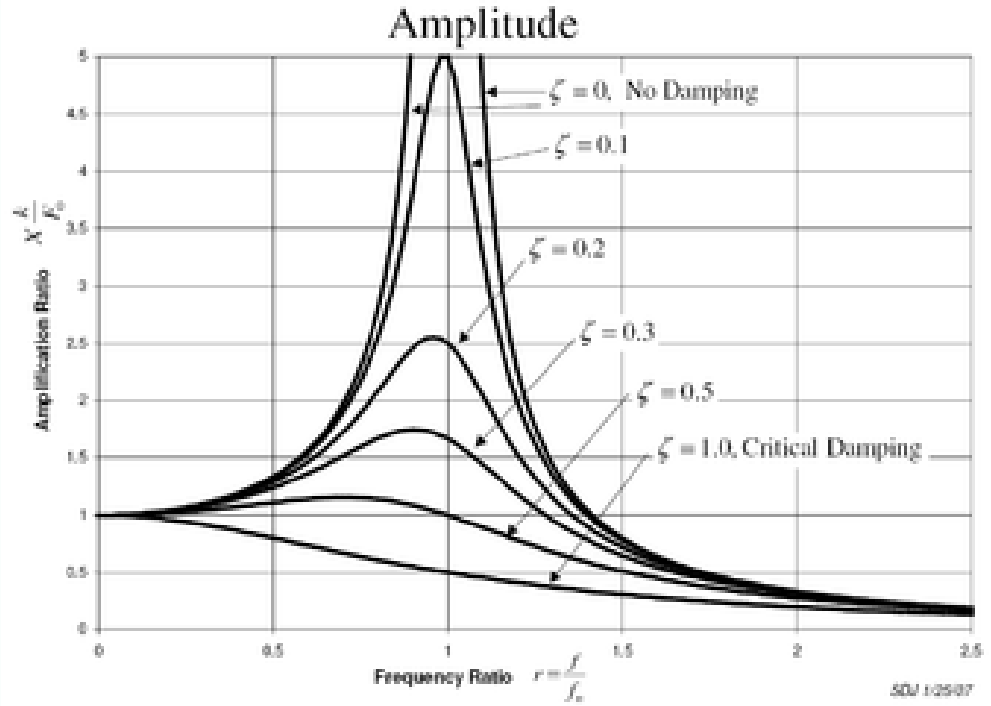
$$@\ \tau = 1, M = \frac{1}{2\xi} = Q @\ \xi = 0\ \ \&\ M = 1; \tau = 0\ or\ \sqrt{2}$$

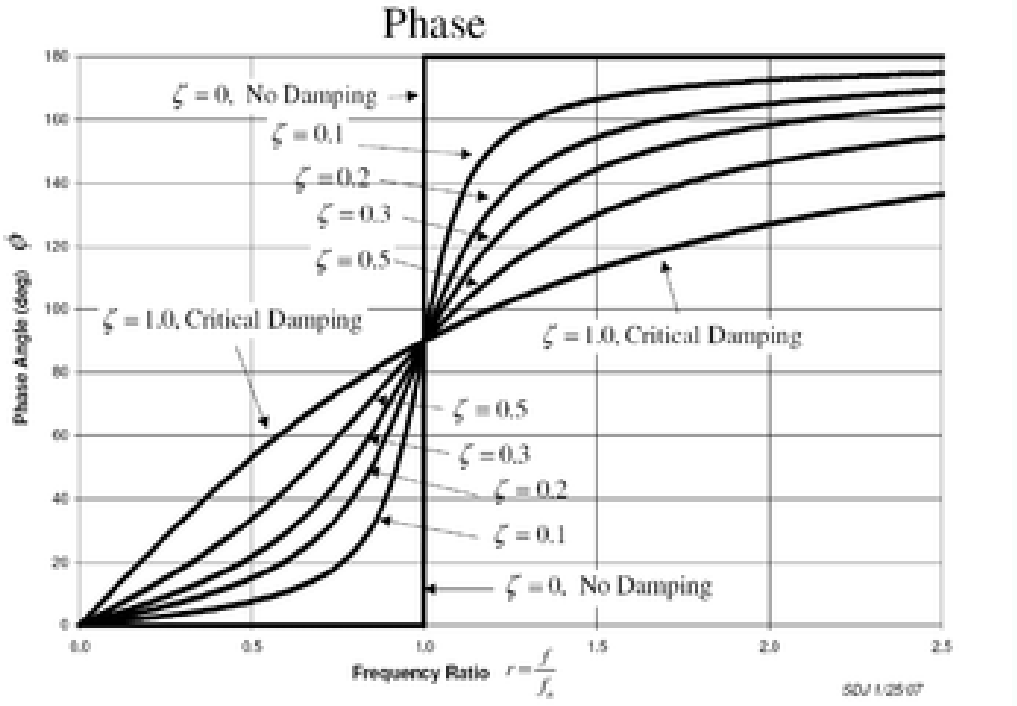
$$\emptyset = \tan^{-1}\frac{(2\xi\tau)}{(1-\tau^2)}$$

$$@\ \tau = 0, \emptyset = 0^o$$

$$@\ \tau = 1, \emptyset = 90^o$$

$$\text{as}\ \tau \to \infty, \emptyset \to 180^o$$





Example 2–3

Find the total response of a single-degree-of-freedom system with m=10 kg, c=20 kg/s and k= 4 kN/m, x(0)=0.01 m,

$\dot{x}(0) = 0, F(t) = 100\cos 10t\ N$

Solution

$$c_c = 2\sqrt{mk} = 2\sqrt{40\,000} = 400\ kg/s$$

$$\xi = \frac{c}{c_c} = 0.05$$

$$\omega_n = \sqrt{\frac{k}{m}} = 20\ rad/s$$

$$\omega_d = \omega_n\sqrt{1-\xi^2} = 19.97\ rad/s$$

$$\tau = \frac{\omega}{\omega_n} = \frac{10}{20} = 0.5$$

$$\delta = \frac{F_o}{k} = \frac{100}{4000} = 0.025\ m$$

$$X_o = \frac{\delta}{\sqrt{(1-\tau^2)^2 + (2\xi\tau)^2}} = \frac{0.025}{\sqrt{(0.75)^2 + (0.05)^2}}$$

$$= 33.26\ mm$$

$$\xi\omega_n = 1$$

$$x = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi) + X_o\cos(\omega t - \theta)$$

$$\dot{x} = A\omega_d e^{-t}\cos(\omega_d t + \phi) - Ae^{-t}\sin(\omega_d t + \phi) - X_o\,\omega\sin(\omega t - \theta)$$

$$\theta = \tan^{-1}\left(\frac{2\xi\tau}{1-\tau^2}\right) = \tan^{-1} 0.067 = 3.81°$$

Example 2-3

$$x = Ae^{-t}\sin(\omega_d t + \phi) + X_o \cos(\omega t - \theta)$$
$$\dot{x} = A\omega_d e^{-t}\cos(\omega_d t + \phi) - Ae^{-t}\sin(\omega_d t + \phi) - X_o\,\omega \sin(\omega t - \theta)$$

Applying initial conditions

$$x(0) = 10 = A\sin\phi + 33.19$$
$$A\sin\phi = -23.19 \ldots\ldots 1$$
$$0 = v(0) = \mathrm{A}\omega_d\cos(\phi) - A\sin(\phi) - X_o\,\omega\sin(-\theta)$$
$$A\cos\phi = -2.27 \ldots\ldots \ 2$$

Solving 1 and 2 simultaneously gives

$$A = 23.30\,mm \quad and \quad \emptyset = 264.41°$$
$$x(t) = 23.3e^{-t}\sin(19.97t + 264.41°) + 33.26\cos(10t + 3.81)\,mm$$

Quality Factor and Bandwidth

- The value of the amplitude ratio at resonance is also called *Q factor* or *quality factor* of a system

$$@\ \tau = 1, M = \frac{1}{2\xi} = Q$$

- The points R_1 and R_2 where the amplification factor falls to $Q/\sqrt{2}$ are called *halfpower points.*

$$\frac{1}{\sqrt{(1-\tau^2)^2 + (2\xi\tau)^2}} = \frac{Q}{\sqrt{2}} = \frac{1}{2\xi\sqrt{2}}$$

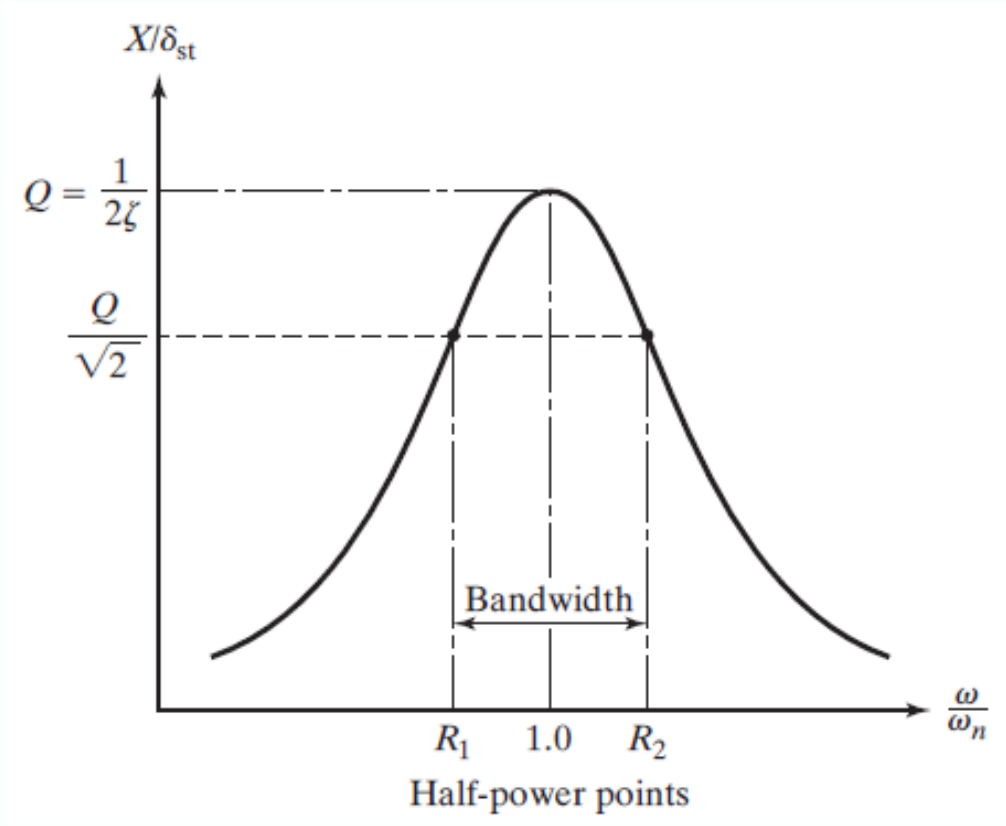
For small values of ξ

$$\tau_1 = R_1 = \frac{\omega_1}{\omega_n} = \sqrt{1-2\xi} \quad \text{and } \tau_1 = R_2 = \frac{\omega_2}{\omega_n} = \sqrt{1+2\xi}$$

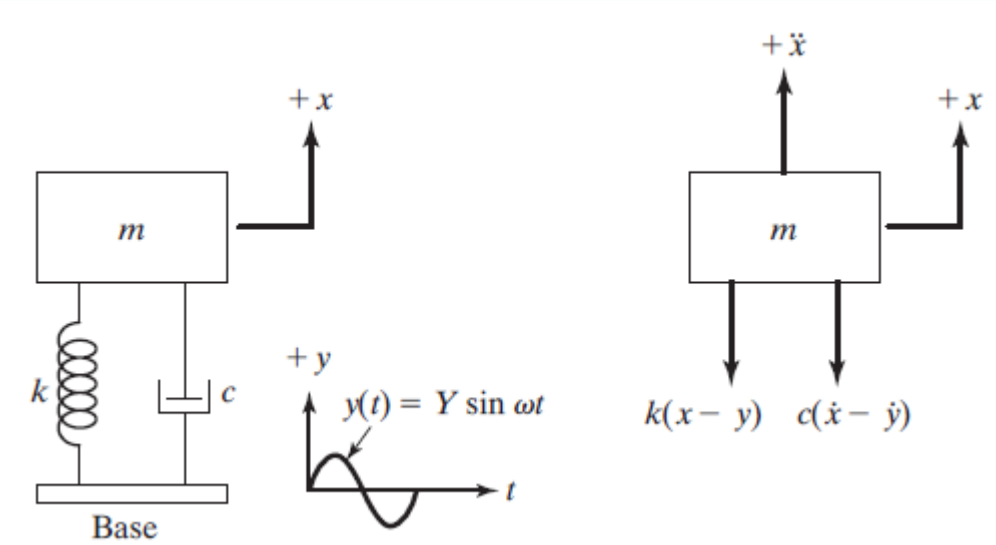
$$\omega_n = \frac{\omega_1 + \omega_2}{2};\ {\omega_2}^2 - {\omega_1}^2 = 4\xi{\omega_n}^2;$$

$$\Delta\omega = \omega_2 - \omega_1 = 2\xi\omega_n$$

$$\therefore Q = \frac{1}{2\xi} = \frac{\omega_n}{\Delta\omega}$$



Base Excitation



EQUATION OF MOTION

$$m\ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0$$

$$y(t) = Y\,sin\omega\,t$$

ABSOLUTE DISPLACEMENT, X

$$m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$$

$$Amplitude\ ratio = T$$

$$T = \frac{X}{Y} = \sqrt{\frac{1 + (2\xi\tau)^2}{(1 - \tau^2)^2 + (2\xi\tau)^2}}$$

RELATIVE DISPLACEMENT, z

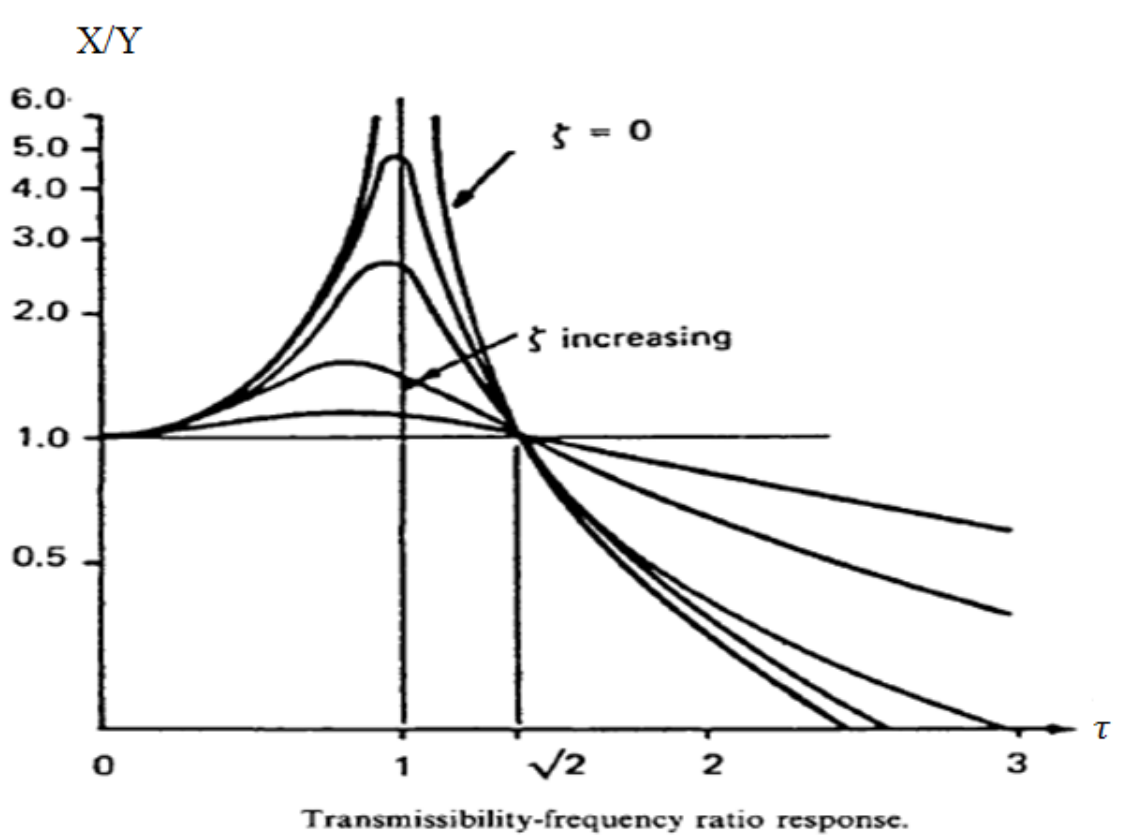
$$let\ z = x - y$$

$$m\ddot{z} + c\dot{z} + kz = -m\ddot{y}$$

$$\frac{Z}{Y} = \frac{\tau^2}{\sqrt{(1 - \tau^2)^2 + (2\xi\tau)^2}}$$

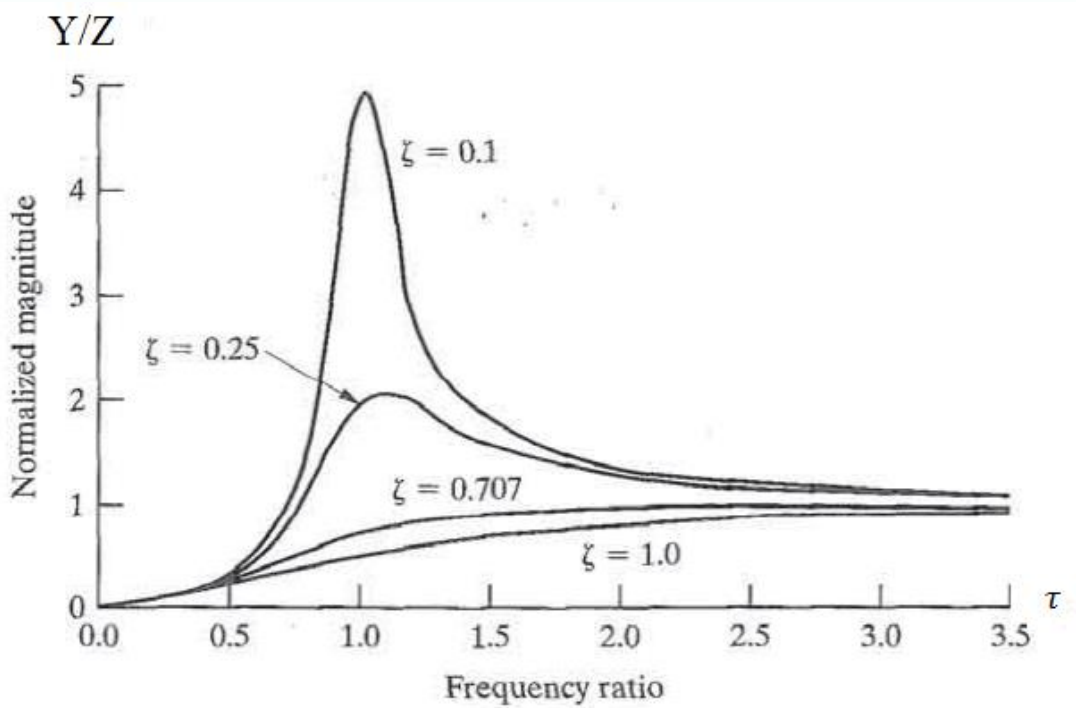
Base Excitation

- ABSOLUTE DISPLACEMENT RESPONSE



Transmissibility-frequency ratio response.

RELATIVE DISPLACEMENT RESPONSE

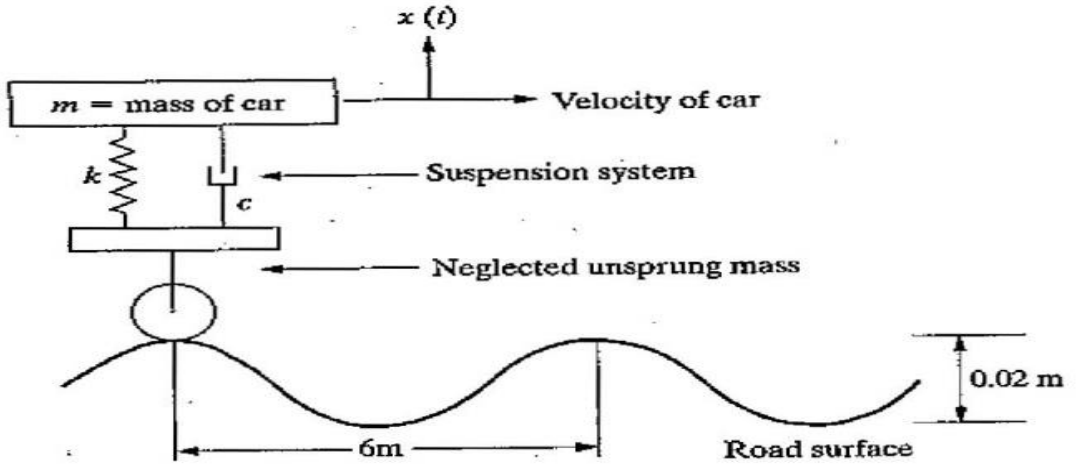


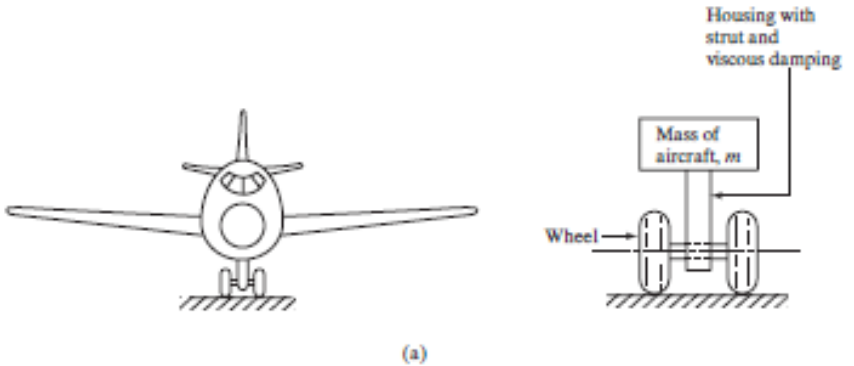
Example 6

A very common example of base motion is the single-degree-of-freedom model of an automobile driving over a rough road or an airplane taxiing over a rough runway, indicated below. The road is approximated as sinusoidal in cross section providing a base motion displacement of

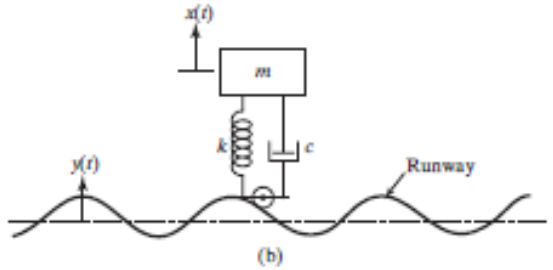
$$y(t) = 0.01 \sin \omega\, t \text{ m}$$

Assume a vehicle of mass 1007 kg and stiffness 40 kN/m is moving with a speed of 20 km/h on such a road. Calculate the vertical displacement of the vehicle.





(a)



(b)

Solution

$$\omega = \frac{2\pi v}{\lambda} = 5.818 \text{ rad/s}$$

where *v* denotes the vehicle's velocity in m/s.

Solution

$$\omega_n = \sqrt{\frac{4 \times 10^5 \text{ N/m}}{1007 \text{ kg}}} = 19.93 \text{ rad/s}$$

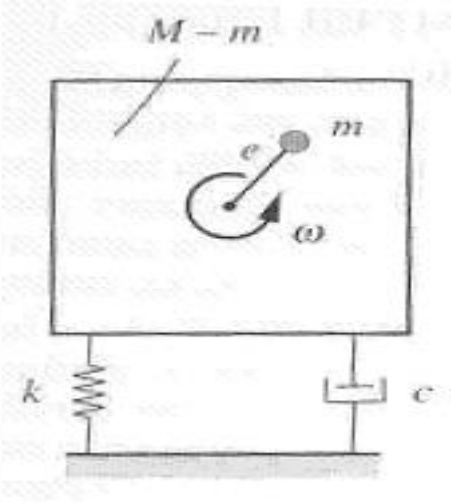
So that τ = 5.818/19.93 = 0.292 and

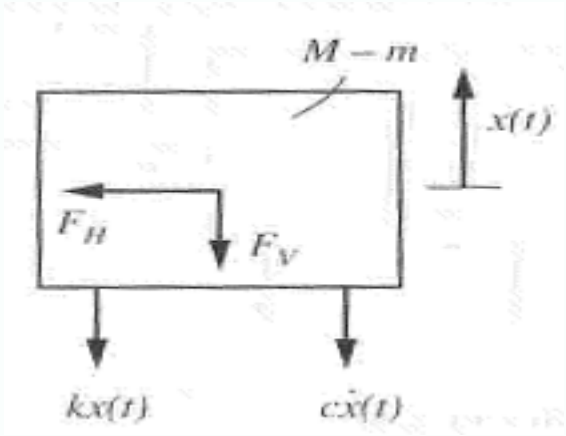
$$\xi = \frac{c}{2\sqrt{km}} = \frac{20{,}000 \text{ N.s/m}}{2(4 \times 10^5 \text{ N/m})(1007 \text{ kg})} = 0.498$$

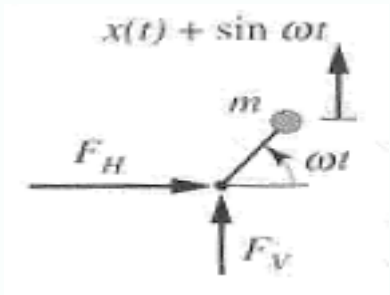
The deflection experienced by the car will be

$$X = (0.01 \text{ m}) \sqrt{\frac{1 + [2(0.498)(0.292)]^2}{[1 - (0.292)^2]^2 + [2(0.498)(0.292)]^2}} = 0.0108 \text{ m}$$

Rotating Unbalance







Displacement of Total Mass

$$(\mathrm{M}-\mathrm{m})\ddot{\mathrm{x}}+\mathrm{m}\frac{d^2}{dt^2}(x+esin\omega t)=-\mathrm{c}\dot{\mathrm{x}}-\mathrm{kx}$$

$$\mathrm{M}\ddot{\mathrm{x}}+\mathrm{c}\dot{\mathrm{x}}+\mathrm{kx}=\mathrm{m}\omega^2\, e\,\sin\omega\,\mathrm{t}$$

$$X=\frac{me}{M}\frac{\tau^2}{\sqrt{(1-\tau^2)^2+(2\xi\tau)^2}}$$

Force Transmitted to support

$$F_T=\mathrm{c}\dot{\mathrm{x}}+\mathrm{kx}$$

$$x=X_o\sin(\omega\,\mathrm{t}-\emptyset)$$

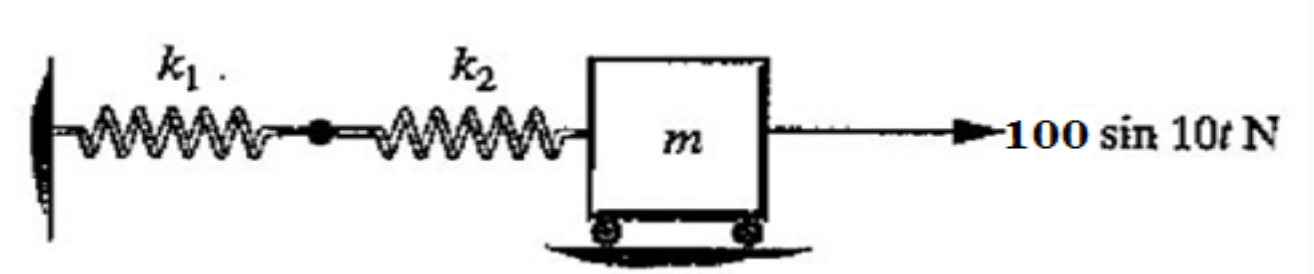
$$F_T=F_o\sqrt{\frac{1+(2\xi\tau)^2}{(1-\tau^2)^2+(2\xi\tau)^2}}$$

$$Transmissibility=T=\frac{F_T}{F_o}$$

$$\emptyset=\tan^{-1}\frac{(2\xi\tau)}{(1-\tau^2)}$$

Assignment 1

Consider the system in the Figure below. Write the equation of motion, and calculate the response assuming that the system is initially at rest for the values k_1, = 1 kN/m, k_2= 5 kN/m, and m = 50 kg.



Plot the response of this system using **excel**

ASSIGNMENT 2

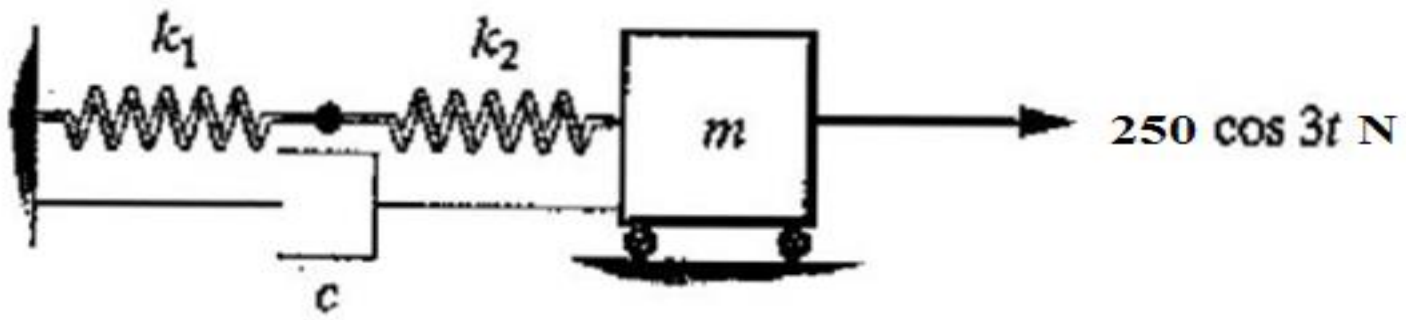
A generator of mass 100 kg is supported on springs which deflect 20 mm under the load. The vibration of the machine on the springs is constrained to be linear and vertical, and a dash-pot fitted in order to reduce the amplitude to one-quarter of its initial value in two complete oscillations. If a sinusoidal force of magnitude 50 N is applied at 10 Hz.

Find the maximum amplitude gain, bandwidth and quality factor of the generator.

Assignment 3

Compute the response of the system below, if the system is initially at x(0)=0.10m and v(0)=-0.02 m/s

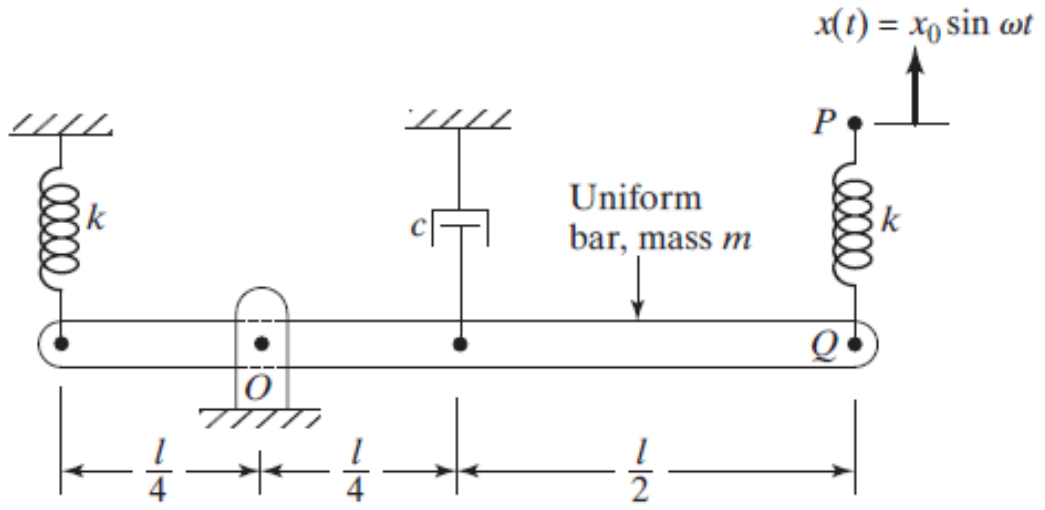
k_1 = 1 k N/m, k_2 = 5 kN/m, c = 200 kg/s, and m = 50 kg.



Plot the response of this system using Microsoft **Excel**

Assignment 4

- A uniform bar of mass *m* is pivoted at point *O* and supported at the ends by two springs, as shown. End *P* of spring *PQ* is subjected to a sinusoidal displacement
- Find the steady-state angular displacement of the bar when l=1m, k = 1000 N/m, c = 500 N-s/m, m = 10 kg, xo=1cm and ω = 10 rad/s.



Assignment 5

One of the tail rotor blades of a helicopter has an unbalanced mass of m=0.5 kg at a distance of e=0.15 m from the axis of rotation, as shown. The tail section has a length of 4 m, a mass of 240 kg, a flexural stiffness (*EI*) of 2.5 MNm2 and a damping ratio of 0.15. The mass of the tail rotor blades, including their drive system, is 20 kg. Determine the forced response of the tail section when the blades rotate at 1500 rpm.

